

UpGen: Unleashing Potential of Foundation Models for Training-Free Camouflage Detection via Generative Models

Ji Du¹, Jiesheng Wu¹, Desheng Kong, Weiyun Liang¹, Fangwei Hao¹, Jing Xu¹, *Member, IEEE*, Bin Wang¹, Guiling Wang¹, *Fellow, IEEE*, and Ping Li², *Member, IEEE*

Abstract—Camouflaged Object Detection (COD) aims to segment objects resembling their environment. To address the challenges of extensive annotations and complex optimizations in supervised learning, recent prompt-based segmentation methods excavate insightful prompts from Large Vision-Language Models (LVLMs) and refine them using various foundation models. These are subsequently fed into the Segment Anything Model (SAM) for segmentation. However, due to the hallucinations of LVLMs and insufficient image-prompt interactions during the refinement stage, these prompts often struggle to capture well-established class differentiation and localization of camouflaged objects, resulting in performance degradation. To provide SAM with more informative prompts, we present UpGen, a pipeline that prompts SAM with generative prompts without requiring training, marking a novel integration of generative models with LVLMs. Specifically, we propose the Multi-Student-Single-Teacher (MSST) knowledge integration framework to alleviate hallucinations of LVLMs. This framework integrates insights from multiple sources to enhance the classification of camouflaged objects. To enhance interactions during the prompt refinement stage, we are the first to leverage generative models on real camouflage images to produce SAM-style prompts without fine-tuning. By capitalizing on the unique learning mechanism and structure of generative models, we effectively enable image-prompt interactions and generate highly informative prompts for SAM. Our extensive experiments demonstrate that UpGen outperforms weakly-supervised models and its SAM-based counterparts. We also integrate our framework into existing weakly-supervised methods to generate pseudo-labels,

resulting in consistent performance gains. Moreover, with minor adjustments, UpGen shows promising results in open-vocabulary COD, referring COD, salient object detection, marine animal segmentation, and transparent object segmentation.

Index Terms—COD, prompt-based segmentation, SAM, large vision-language models, generative models.

I. INTRODUCTION

CAMOUFLAGED Object Detection (COD) aims to break down camouflage and segment objects closely resembling their environment [1]. Most efforts have targeted on fully supervised learning with meticulously crafted modules [2], [3], [4], extra information [5], [6], [7], [8], or novel learning paradigms [9], [10], [11], [12]. To alleviate reliance on time-consuming and labor-intensive data annotation, weakly supervised COD aims to learn semantic-rich and structural information from scribbles [13] or pseudo masks [14]. However, manual annotations remain a necessity in this line of research.

With the advent of vision foundation models like SAM [15], [16], [17], [18], pioneering attempts [19], [20] in leveraging foundation models for COD have ignited the community's enthusiasm for embracing large models. Subsequent prompt-based segmentation methods [21], [22], [23], [24] have made significant progress on the way, mainly focusing on generating different types of prompts via LVLMs (e.g. coordinates [25], text [26]), and fed them into SAM for segmentation. Nevertheless, these methods either suffer from inaccurate coordinates or require fine-tuning on SAM. To remedy the hallucination [27], [28], [29] (LVLMs produce incorrect statements to the input images and prompts), compared to getting “hard” prompts (“hard” means prompts from LVLMs are directly input to SAM without any interaction with images), several methods concentrate on deriving “soft” prompts (e.g. image-level categories, coarse bounding boxes) and enhance interactions between images and those soft prompts. MMCPF [23] enhances initial coarse bounding boxes with DINOv2 [30]. GenSAM [31] improves Clip Surgery [32] to obtain class activation maps (CAM) [33], which are used to generate point prompts. Nevertheless, these approaches, either necessitate LVLMs to perform localization that they are not designed for or rely on CLIP for prompt refinement, which struggles on pixel-level dense prediction tasks [34], [35], [36], [37]. Inspired by the aforementioned efforts, we observe that the

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Ji Du is with the Department of Computing, The Hong Kong Polytechnic University, Hong Kong, and also with the College of Artificial Intelligence, Nankai University, Tianjin 300071, China (e-mail: polyudui.du@connect.polyu.hk).

Jiesheng Wu, Desheng Kong, Weiyun Liang, Fangwei Hao, Jing Xu, and Bin Wang are with the College of Artificial Intelligence, Nankai University, Tianjin 300071, China (e-mail: jasonwu@mail.nankai.edu.cn; kongds@mail.nankai.edu.cn; weiyunliang@mail.nankai.edu.cn; haofangwei@mail.nankai.edu.cn; xujing@nankai.edu.cn; wb@nankai.edu.cn).

Guiling Wang is with the Department of Computer Science, New Jersey Institute of Technology, Newark, NJ 07102 USA (e-mail: gwang@njit.edu).

Ping Li is with the Department of Computing and Research Institute for Sports Science and Technology, The Hong Kong Polytechnic University, Hong Kong (e-mail: p.li@polyu.edu.hk).

Data is available on-line at <https://github.com/xiaohainku/UpGen>
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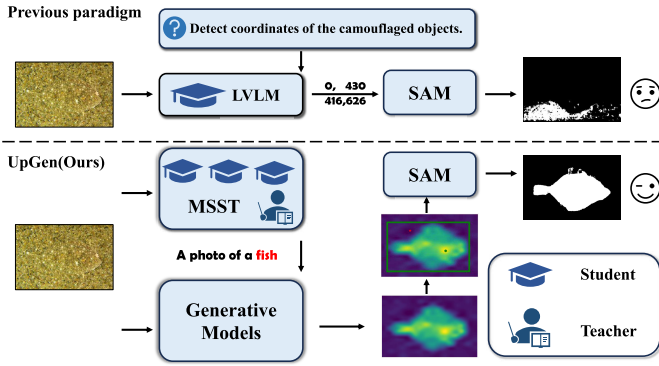


Fig. 1. Comparison of UpGen with previous paradigms. The hallucination of LVLms poses significant challenges to directly generating coordinates. Our approach aims to allow LVLms to generate coarse image-level prompts, which are used for fine-grained prompts via generative models.

following two points are essential for prompt-based segmentation: ① “soft” prompts mean less hallucination of LVLms than “hard” prompts; ② effectively facilitating interactions between images and prompts from LVLms could generate finer-grained prompts, achieving well-established class discrimination and localization.

In this study, we propose UpGen, a training-free pipeline to prompt SAM with generative prompts, as exemplified in Fig. 1. The key idea is to leverage the powerful conditional generation capability of prevalent generative models (e.g. Stable Diffusion [38], Imagen [39], DALLÉ-2 [40]) to enhance interactions between images and categories from LVLms. While deriving soft prompts from LVLms reduces the risk of hallucinations, the inherently challenging properties of COD still present a substantial risk of misclassification for camouflaged objects by LVLms. To further alleviate the hallucination of LVLms and obtain more accurate object categories as soft prompts, we propose the Multi-Student-Single-Teacher framework (MSST). In MSST, multiple students generate knowledge from different views, which is summarized and reasoned by a teacher to obtain object categories. Specifically, we instantiate three independent MSST variants: a) LVLms as students and Large Language Model (LLM) as the teacher, b) prompts as students and voting as the teacher, and c) LVLms as students and CLIP as the teacher.

Soft prompts from MSST primarily offer image-level category information. To achieve more precise object localization, we need to improve the interactions between prompts and images. Generative models, trained on web-scale data, have shown significant potential in data synthesis and image editing. However, their application on real images, especially challenging camouflaged images, remains largely unexplored. In this paper, we empirically demonstrate that without any fine-tuning, generative models could still capture semantic information embedded in COD images. With this observation, we leverage off-the-shelf generative models to enhance the interplay between images and soft prompts from MSST, generating cross-attention maps that effectively indicate object localization and class discrimination. These attention maps are then used to identify high-confidence points and bounding boxes to guide the segmentation process in SAM.

On COD10k [1], UpGen outperforms weakly supervised model [13] by 8.7% and the LVLm-based model [23] by 7.3% in S_α [41]. Meanwhile, with minimal modifications to MSST prompts, UpGen achieves promising results in transparent object segmentation [42], salient object detection [43] and marine animal segmentation [44]. Moreover, our framework could be incorporated into existing weakly-supervised methods, bringing improved performance through generating pseudo-labels.

In summary, our main contributions are as follows:

- We propose UpGen, a training-free pipeline leveraging generative models to conjugate vision and multi-modal foundation models to solve the challenging COD task. To the best of our knowledge, this is the first paradigm to integrate the capabilities of different foundation models into downstream tasks through generative models.
- To alleviate the hallucination of LVLms, we propose MSST, a framework synthesizing knowledge from different aspects for more accurate camouflaged object recognition. To showcase the superiority and versatility of MSST, we instantiate three different variants, varying in students and teachers.
- To enhance interactions between images and knowledge from MSST, we propose to capitalize on the off-the-shelf generative models to query crucial information between real COD images and MSST output.
- Experiments have shown that UpGen significantly outperforms weakly supervised models and its counterparts, setting a solid baseline for COD with generative prompts. Meanwhile, our framework could be integrated into existing weakly supervised methods to boost performance. In addition, UpGen demonstrates promising potential in various downstream tasks with minor modifications.

II. RELATED WORK

A. Camouflaged Object Detection

The key idea of COD is to segment objects that exhibit similar patterns to their environment [1], [2]. Community endeavors mainly focus on designing novel modules [1], [2], [3], [45], [46], [47], [48] and adding auxiliary tasks [5], [6], [7], [49], [50], [51], [52]. In addition, the notion of camouflage as opposed to detection has also spawned the application of the adversarial learning paradigm to COD [9], [51], [53]. Meanwhile, much of the work draws inspiration from a biological perspective [4], [54], [55]. Despite the progress, these methods rely heavily on painstakingly labeled datasets. For example, a single image takes roughly 60 minutes to label, making it incredibly laborious to craft a large-scale dataset [1].

In this paper, we aim at segmenting camouflaged objects without any manual annotations. Our pipeline can be utilized for generating pseudo-labels, reducing labelling time from ~ 1 hour to ~ 1 second.

B. Large Vision-Language Models

Trained on web-scale image-text pairs [56], [57], [58], LVLms [59], [60], [61], [62], [63], [64] highlight increasing

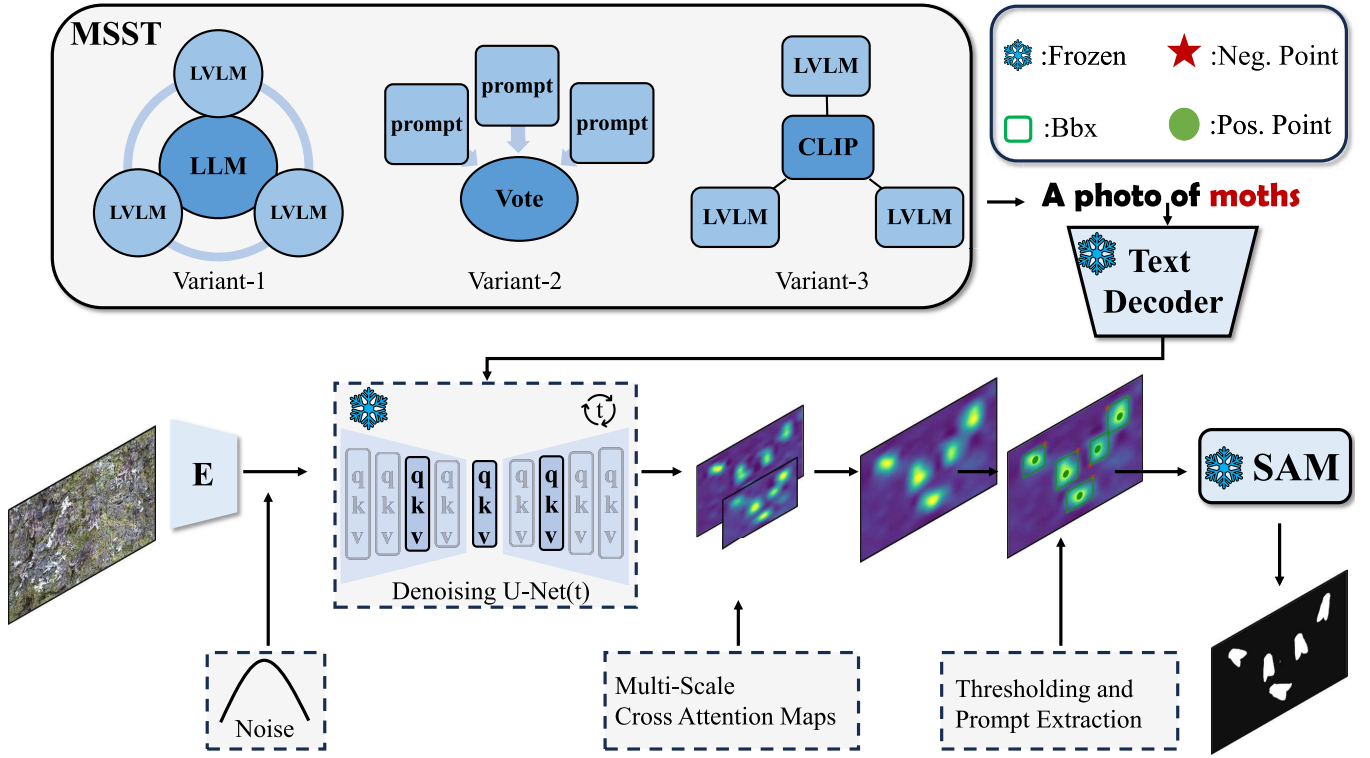


Fig. 2. The overall pipeline of the proposed UpGen consists of three main stages. The first stage instantiates an MSST variant to obtain categories of camouflaged objects. In the second stage, the diffusion model receives knowledge from MSST and acquires multi-scale and multi-time-step attention maps through the cross-attention mechanism. In the last step, prompts are extracted through a thresholding approach to prompt SAM for segmentation.

importance on vision tasks [65], [66], [67], [68]. Previous studies primarily utilize LVLMs to obtain coordinates [23], [25] and categories [22], [26] of camouflaged objects through prompt engineering. Despite the strong relevance of these well-crafted prompts to COD, LVLMs usually fail to provide satisfactory results due to hallucination [27], [28], [29]. To this end, we design the MSST framework, which allows for the effective integration of knowledge from diverse students, such as different prompts or LVLMs.

C. Generative Models

As a branch of generative models, diffusion models [38], [39], [40], [69], [70] achieve unparalleled success in fields such as image generation [71] and image editing [72]. Recent studies also demonstrate the effectiveness of class discrimination and localization of attention maps from diffusion models and use them for annotation generation [73], [74], localization [31] and segmentation [75]. DiffMask [73] proposes to synthesize images using a diffusion model and extract cross-attention maps as annotations, but it can be only applied to synthesized images. DiffSeg [75] extracts self-attention maps for unsupervised zero-shot segmentation, but its segmentation results cannot provide explicit semantic information. Inspired by these efforts, we adopt diffusion models to enhance the interactions between real images and categories from MSST. Previous studies in COD also leverage diffusion models for data augmentation [76] or camouflaged image generation [77]. We are the first to utilize diffusion models for efficient image-text interactions, excavating semantic-rich prompts for COD.

III. METHODOLOGY

We introduce a novel pipeline termed UpGen to prompt SAM with generative prompts. As illustrated in Fig. 2, our UpGen consists of three main stages. Given an image $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$ and a task-related prompt P , MSST is first adopted to get the category C of the camouflaged object. Interactions between $\mathbf{I}$ and C are then enhanced in denoising UNet of Stable Diffusion. Based on that, cross-attention maps $\mathbf{A}$ are extracted for generating points p and bounding boxes b to prompt SAM:

$$C = \text{MSST}(\mathbf{I}, P), \quad (1)$$

$$\mathbf{A} = \text{CrossAttn}(\text{Diffu}(\mathcal{E}(\mathbf{I})), \tau_\theta(C)), \quad (2)$$

$$\{p, b\} = \text{Threshold}(\mathbf{A}), \quad (3)$$

$$\text{Seg} = \text{SAM}(\mathbf{I}, \{p, b\}), \quad (4)$$

where CrossAttn, Diffu, $\mathcal{E}$, τ_θ , Threshold denote cross attention layers of UNet, diffusion process, encoder of variational autoencoder, text encoder, and prompt extraction, respectively.

A. Multi-Student-Single-Teacher Framework

By connecting multiple students through a single teacher, our proposed MSST architecture aims to effectively mitigate the potential hallucinations faced by individual students. To validate the efficacy of this architecture, we instantiated three variants comprising different student-teacher configurations, all of which **operate independently**. We hope these variants could pave the way for further exploration of more effective variants.

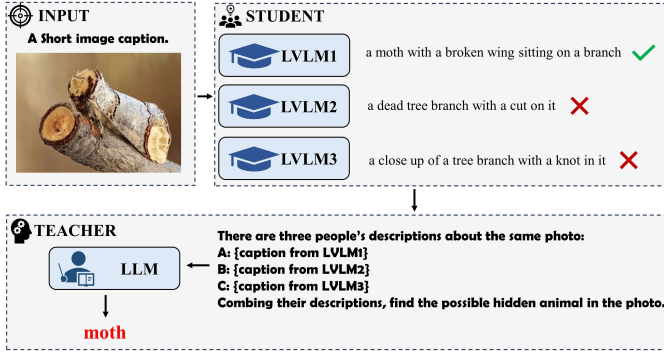


Fig. 3. MSST Variant-1: LVLMs act as the students and LLM as the teacher.

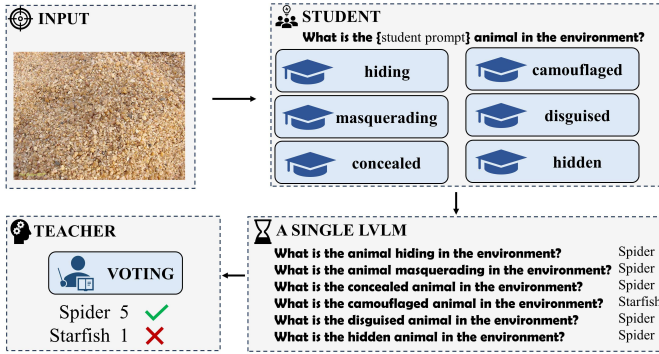


Fig. 4. MSST Variant-2: Prompts act as the students and LVL1 as the teacher.

1) *Variant-1 (LVL1s-LLM)*: To explore whether COD could be realized through weak task-related prompts, we propose the first variant of MSST: LVL1s-LLM, where multiple LVL1s act as students and an LLM acts as the teacher. As shown in Fig. 3, instead of providing LVL1s with task-relevant prompts such as “Find camouflaged objects”, we simply provide a generic prompt “A short image caption”. Taking into account the possible hallucination of a single model, we introduce three LVL1s to generate descriptions [60], [61], [62]. Meanwhile, inspired by the powerful summarization and reasoning capabilities of LLMs [78], [79], [80], [81], we adopt the LLM [80] as the teacher to refine descriptions from multiple LVL1s. This process could be formulated as

$$\text{captions} = \{\text{LVL1}_1(\mathbf{I}, P_s), \text{LVL1}_2(\mathbf{I}, P_s), \text{LVL1}_3(\mathbf{I}, P_s)\}, \quad (5)$$

$$C = \text{LLM}(\text{captions}, P_t). \quad (6)$$

P_s and P_t represent prompts required by LVL1s and LLM, respectively.

2) *Variant-2 (Prompts-Voting)*: To further exploit the potential of a single LVL1, we propose the second MSST variant: Prompts-Voting, where prompts act as students and the teacher is responsible for voting. For a single LVL1, we design a variety of prompts, as shown in Fig. 4. Different task-related prompts elicit the LVL1 to understand COD from different perspectives, avoiding possible understanding deviations that a single prompt may cause. A simple voting mechanism

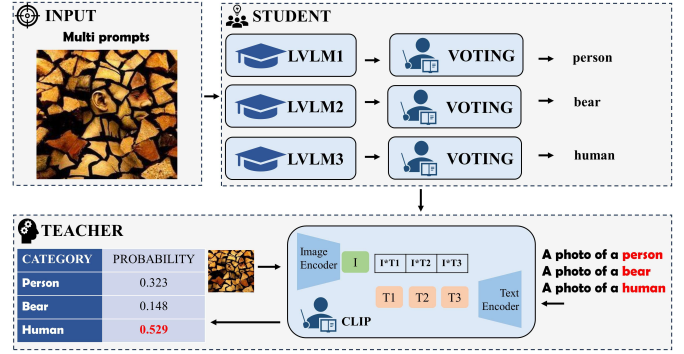


Fig. 5. MSST Variant-3: LVL1s act as the students and CLIP as the teacher.

is adopted to get the final result. Given a set of prompts $P_1, P_2, \dots, P_n$, we formulate this process as

$$\text{categories} = \text{LVL1}(P_1, P_2, \dots, P_n), \quad (7)$$

$$C = \text{Voting}(\text{categories}). \quad (8)$$

3) *Variant-3 (LVL1s-CLIP)*: The third variant LVL1s-CLIP is proposed to utilize multiple LVL1s, where LVL1s act as students and CLIP acts as the teacher, as shown in Fig. 5. Specifically, output from different LVL1s is regarded as the possible GT category, which are input to CLIP for performing classification to get the final category. Given an image $\mathbf{I}$ and a set of categories S , their representations in embedding space could be computed by

$$\mathbf{I}_{\text{rep}} = \text{Proj}(\text{CLIP_image_encoder}(\mathbf{I}), \mathbf{W}_i), \quad (9)$$

$$\mathbf{S}_{\text{rep}} = \text{Proj}(\text{CLIP_text_encoder}(S), \mathbf{W}_t). \quad (10)$$

$\mathbf{W}_i$ and $\mathbf{W}_t$ represent the learned projection of image and text features to embedding space, respectively. The final category is determined by the max probability:

$$\text{probs} = \text{Softmax}(\text{dot}(\mathbf{I}_{\text{rep}}, \mathbf{S}_{\text{rep}})), \quad (11)$$

$$C = S[\arg \max(\text{probs})]. \quad (12)$$

4) *Clarifications*: It is important to note that our MSST is orthogonal to the general-purpose approaches aimed at reducing hallucinations. While these methods typically address hallucinations in individual LVL1s by fine-tuning [82], [83] or enhancing decoding strategies [84], [85], MSST focuses on overcoming challenges faced by generic LVL1s in specific downstream tasks through knowledge fusion. Together, these approaches can complement each other, ultimately enhancing the reliability of LVL1s.

B. Generative Prompts

Based on the categories from MSST, we adopt generative models to obtain the finer-grained information, such as localization and class-differentiation of camouflaged objects.

1) *Formulation*: We adopt Stable Diffusion [38] as the generative model. Specifically, Stable Diffusion consists of three components: a text encoder τ_θ , a variational autoencoder (VAE), and a time-conditional UNet. Given an image $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$ in RGB space, the VAE encoder $\mathcal{E}$ encodes $\mathbf{I}$ into

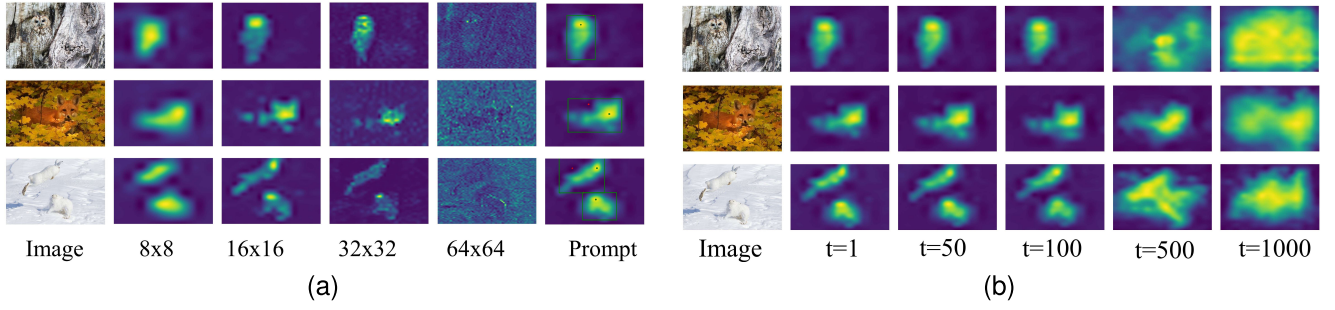


Fig. 6. Attention maps with (a) different resolutions, and (b) different time steps.

latent representation space as $\mathbf{z} = \mathcal{E}(\mathbf{I})$. The diffusion process at time step t can be formulated as

$$\mathbf{z}_t \triangleq \sqrt{\alpha_t} \mathbf{z} + \sqrt{1 - \alpha_t} \boldsymbol{\epsilon}, \boldsymbol{\epsilon} \sim \mathcal{N}(0, 1), \quad (13)$$

$\alpha_1, \dots, \alpha_T$ represent the noise schedule and $\bar{\alpha}_t = \prod_{k=1}^t \alpha_k$, as defined in [69]. During denoising, the attention U-Net is optimized by

$$L = \mathbb{E}_{\mathbf{z}, y, \boldsymbol{\epsilon}, t} [\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t, \tau_\theta(y))\|_2^2], \quad (14)$$

where $\boldsymbol{\epsilon}_\theta$ denotes the U-Net, conditioned on time step t and encoded prompt embedding $\tau_\theta(y) \in \mathbb{R}^{N \times d_\tau}$ (y , N , and d_τ denote the Stable Diffusion prompt, the token length, and the intermediate representation dimension, respectively). In this paper, we simply set y to “a photo of a [class]”.

2) *Extract Cross-Attention Maps*: The UNet of Stable Diffusion consists of three main components: the encoder, the bottleneck, and the decoder. Both the encoder and decoder comprise four stages: three cross-attention blocks at resolutions of $\times 64$, $\times 32$, and $\times 16$, followed by one block (without cross-attention layers) at a resolution of $\times 8$. The bottleneck, which includes cross-attention layers, operates exclusively at a resolution of $\times 8$. At each time step, text and image interactions occur at cross-attention layers of different stages. Formally, cross attention maps at time step t with spatial resolution $r \in \{\times 8, \times 16, \times 32, \times 64\}$ could be derived by

$$\mathbf{A}_{r,t} = \text{Softmax}(\mathbf{Q}\mathbf{K}^\top / \sqrt{d}), \quad (15)$$

$$\mathbf{Q} = \mathbf{W}_Q \boldsymbol{\varphi}(\mathbf{z}_t), \mathbf{K} = \mathbf{W}_K \tau_\theta(y), \quad (16)$$

where $\mathbf{W}_Q$ and $\mathbf{W}_K$ refer to learned projections, $\boldsymbol{\varphi}(\mathbf{z}_t)$ denotes the flattened intermediate representation of $\mathbf{z}_t$. After reshaping, $\mathbf{A}_{r,t} \in \mathbb{R}^{H \times W \times N}$ represents the cross-attention maps of N textual tokens. When extracting multi-scale and multi-time-step cross-attention maps, we only consider the textual token corresponding to the categories. The average attention map can be computed as

$$\mathbf{A} = 1/(|R||T|) \sum_{r \in R, t \in T} \text{Normalize}(\mathbf{A}_{r,t}[:, :, c]), \quad (17)$$

where R , T and c refer to the set of spatial resolutions, time steps, and the index of the category token, respectively.

It is noteworthy that different cross-attention layers of U-Net contain information at different semantic levels, as shown in Fig. 6(a). Low-resolution maps capture global contextual information, while high-resolution attention maps capture local information and low-level features, such as edges and texture.

In this study, attention maps with resolutions of $\times 8$ (from the UNet bottleneck) and $\times 16$ (from the UNet encoder and decoder) are chosen. In addition, different time steps of attention maps contain semantics that match the images differently. A low time step corresponds to a small noise variance, which means less image distortion. In general, the diffusion model is capable of denoising even with large time steps. However, as we do not fine-tune the model, the latent image after the diffusion process can be approximated as noise at large time steps. The denoising process, at this point, is equivalent to the inference process of the model, which ultimately generates a new image based on the prompt. This is the reason why the attention maps generated at time steps 1 and 1,000 are completely different in Fig. 6(b). Experimentally, we choose attention maps at time step 1, 50, and 100.

3) *Extract Prompts*: To generate prompts from attention maps, image-level and bounding-box-level (bbx-level) thresholds are proposed. Specifically, the image-level threshold is first applied to attention maps to get a binary mask. Multiple candidate boxes would be generated by computing the minimum bounding rectangle that encloses the external contours of this binary image. To filter noisy bounding boxes, we consider the box to be valid only if it contains points of high confidence, which is defined by the bbx-level threshold. Furthermore, for the filtered boxes, we sample an equal number of points with the highest and lowest confidence as positive and negative samples, respectively. Experimentally, the image-level threshold, bbx-level threshold, and the number of sampling points are set to 0.3, 0.9, and 1, respectively.

IV. EXPERIMENTS

A. Experimental Settings

We conduct experiments on four widely used COD datasets, including CHAMELEON (76 images) [86], CAMO-TEST (250 images) [87], COD10K-TEST (2,026 images) [1], and NC4K (4,121 images) [52]. Following previous work, mean absolute error (MAE) [88], structure measure (S_a) [41], mean E-measure (E_ϕ) [89] and weighted F-measure (F_β^ω) [90] are adopted to evaluate our method. We choose BLIP2-OPT-6.7B [60], InstructBLIP-Vicuna-7B [62] and LLaVA-1.5-7B [61] as LVLMs and LLaMA-2-7B [80] as the LLM. In Variant 1, all the LVLMs act as students while the LLM as the teacher. In Variant 2, multiple prompts act as students while a single LVM as the teacher. For simplicity, Variant-2-B, Variant-2-I and Variant-2-L stand for BLIP2, InstructBLIP, LLaVa-1.5

TABLE I

QUANTITATIVE COMPARISON WITH SOTAs ON FOUR COD DATASETS. “ $\uparrow$ / $\downarrow$ ”: THE HIGHER/LOWER THE BETTER. “-”: NOT AVAILABLE. “F”, “W” AND “TF” DENOTE “FULLY-SUPERVISED”, “WEAKLY-SUPERVISED” AND “TRAINING-FREE” METHODS, RESPECTIVELY. THE BEST RESULTS OF OTHER METHODS ARE IN BOLD, AND THE TOP TWO RESULTS OF OUR METHODS ARE HIGHLIGHTED IN RED AND GREEN. “+” INDICATES INCORPORATING WEAKLY-SUPERVISED METHODS WITH OUR UPGen FRAMEWORK

Method	Venue	Sup.	CHAMELEON				CAMO				COD10K				NC4K			
			$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^\omega \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^\omega \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^\omega \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^\omega \uparrow$	$M \downarrow$
SINET	CVPR20	F	0.869	0.891	0.740	0.044	0.751	0.771	0.606	0.100	0.771	0.806	0.551	0.051	0.808	0.871	0.723	0.058
C2FNet	IJCAI21	F	0.888	0.935	0.828	0.032	0.796	0.854	0.719	0.080	0.813	0.890	0.686	0.036	0.838	0.897	0.762	0.049
JSCOD	CVPR21	F	0.891	0.945	0.833	0.030	0.800	0.859	0.728	0.073	0.809	0.884	0.684	0.035	0.842	0.898	0.771	0.047
LSR	CVPR21	F	0.890	0.935	0.822	0.030	0.787	0.838	0.696	0.080	0.804	0.880	0.673	0.037	0.840	0.895	0.766	0.048
S-MGL	CVPR21	F	0.892	0.912	0.802	0.032	0.772	0.806	0.664	0.089	0.811	0.844	0.654	0.037	0.829	0.862	0.731	0.055
R-MGL	CVPR21	F	0.893	0.917	0.812	0.031	0.776	0.812	0.673	0.088	0.814	0.851	0.666	0.035	0.833	0.867	0.739	0.053
PFNET	CVPR21	F	0.882	0.931	0.810	0.033	0.782	0.841	0.695	0.085	0.800	0.877	0.660	0.040	0.829	0.887	0.745	0.053
UGTR	ICCV21	F	0.888	0.911	0.796	0.031	0.785	0.823	0.686	0.086	0.818	0.853	0.667	0.035	0.839	0.874	0.747	0.052
PreyNet	MM22	F	0.895	0.952	0.844	0.028	0.790	0.842	0.708	0.077	0.813	0.881	0.697	0.034	0.834	0.887	0.763	0.050
FAPNet	TIP22	F	0.893	0.940	0.825	0.028	0.815	0.865	0.734	0.076	0.822	0.888	0.694	0.036	0.851	0.899	0.775	0.047
BSANet	AAAI22	F	0.895	0.946	0.841	0.027	0.794	0.851	0.717	0.079	0.818	0.891	0.699	0.034	0.841	0.897	0.771	0.048
FDNet	CVPR22	F	0.895	0.951	0.849	0.027	0.840	0.896	0.782	0.063	0.838	0.921	0.747	0.030	0.832	0.895	0.759	0.052
ZoomNet	CVPR22	F	0.902	0.943	0.845	0.023	0.820	0.877	0.752	0.066	0.838	0.888	0.729	0.029	0.853	0.896	0.784	0.043
SegMaR	CVPR22	F	0.906	0.951	0.860	0.025	0.815	0.874	0.753	0.071	0.833	0.899	0.724	0.034	0.841	0.896	0.781	0.046
SINETV2	TPAMI22	F	0.888	0.942	0.816	0.030	0.820	0.882	0.743	0.070	0.815	0.887	0.680	0.037	0.847	0.903	0.770	0.048
HitNet	AAAI23	F	0.921	0.967	0.897	0.019	0.849	0.906	0.809	0.055	0.871	0.935	0.806	0.023	0.875	0.926	0.834	0.037
PopNet	ICCV23	F	0.917	0.965	0.875	0.020	0.808	0.859	0.744	0.077	0.851	0.910	0.757	0.028	0.861	0.909	0.802	0.042
FSNet	TIP23	F	0.905	0.961	0.854	0.023	0.879	0.931	0.838	0.042	0.870	0.936	0.787	0.023	0.891	0.938	0.845	0.031
FSPNet	CVPR23	F	0.908	0.943	0.851	0.023	0.856	0.899	0.799	0.050	0.851	0.895	0.735	0.026	0.879	0.915	0.816	0.035
FEDER	CVPR23	F	0.887	0.946	0.834	0.030	0.802	0.867	0.738	0.071	0.822	0.900	0.716	0.032	0.847	0.907	0.789	0.044
WSCOD	AAAI23	W	0.794	0.870	0.708	0.048	0.731	0.814	0.637	0.095	0.719	0.814	0.552	0.052	0.765	0.847	0.673	0.066
WSCOD+	-	W	0.807	0.898	0.719	0.049	0.742	0.823	0.650	0.099	0.742	0.826	0.581	0.055	0.791	0.861	0.699	0.065
WS-SAM-P	NIPS23	W	0.805	0.868	0.700	0.056	0.718	0.757	0.602	0.102	0.791	0.856	0.663	0.039	0.813	0.859	0.734	0.057
WS-SAM-S	NIPS23	W	0.820	0.887	0.723	0.048	0.759	0.814	0.667	0.092	0.803	0.877	0.680	0.038	0.829	0.886	0.757	0.052
WS-SAM	NIPS23	W	0.805	0.847	0.707	0.077	0.778	0.838	0.684	0.088	0.796	0.862	0.660	0.044	0.832	0.886	0.752	0.054
WS-SAM+	-	W	0.845	0.908	0.771	0.043	0.792	0.861	0.705	0.080	0.797	0.863	0.661	0.042	0.833	0.887	0.751	0.053
MMCPF	MM24	TF	-	-	-	-	0.749	0.820	0.680	0.101	0.733	0.803	0.592	0.066	0.767	0.826	0.681	0.083
GenSAM	AAAI24	TF	0.767	0.827	0.673	0.075	0.738	0.803	0.674	0.106	0.773	0.832	0.667	0.065	0.810	0.866	0.751	0.065
MSST Variant-1 (Ours)	-	TF	0.805	0.851	0.731	0.061	0.774	0.832	0.717	0.095	0.794	0.841	0.693	0.062	0.827	0.873	0.772	0.063
MSST Variant-2-B (Ours)	-	TF	0.814	0.870	0.750	0.057	0.760	0.818	0.690	0.105	0.782	0.827	0.674	0.070	0.818	0.862	0.757	0.070
MSST Variant-2-I (Ours)	-	TF	0.806	0.865	0.738	0.059	0.768	0.816	0.701	0.101	0.804	0.855	0.707	0.052	0.836	0.883	0.785	0.057
MSST Variant-2-L (Ours)	-	TF	0.820	0.874	0.757	0.048	0.793	0.842	0.737	0.089	0.802	0.852	0.703	0.053	0.836	0.883	0.785	0.056
MSST Variant-3 (Ours)	-	TF	0.817	0.875	0.747	0.047	0.779	0.833	0.718	0.091	0.806	0.856	0.708	0.051	0.838	0.887	0.788	0.054

as teachers, respectively. In Variant 3, all the LVLMS act as students while CLIP as the teacher. Being free of training, inference could be run on CPUs or GPUs. We run inference on a single A40 (48G).

B. Comparison With SOTA Methods

1) *Comparison Methods*: To validate the effectiveness of the pipeline UpGen, we conduct comparative experiments with weakly-supervised (WSCOD [13], WS-SAM-P [14] and WS-SAM-S [14]) and training-free methods (MMCPF [23] and GenSAM [22]). We also provide fully-supervised methods for reference, including SINet [1], C2FNet [46], JSCOD [91], LSR [52], S-MGL [92], R-MGL [92], PFNET [45], UGTR [93], PreyNet [94], FAPNet [47], BSANet [49], FDNet [6], ZoomNet [54], SegMaR [55], SINETV2 [2], HitNet [4], PopNet [5], FSNet [48], FSPNet [3], FEDER [95]. All methods are evaluated with the same evaluation setup.

2) *Experimental Results and Analysis*: As shown in Table I, our method UpGen achieves equivalent or better results compared to fully supervised methods before 2022. Notably, on COD10K, MSST Variant-3 outperforms weakly-supervised method WSCOD [13] 8.7% in S_α and 15.6% in F_β^ω , elucidating the effectiveness of the whole pipeline to enable camouflaged object segmentation in a training-free manner. All variants of UpGen substantially outperform training-free SAM-based MMCPF [23] and GenSAM [22] in most metrics, suggesting that the merits of our approach are rooted in the design of MSST knowledge fusion and the innovative design of generative-model-based prompt generation. We also incorporate UpGen into existing weakly-supervised methods

to generate pseudo-labels. The resulting methods, WSCOD+ and WS-SAM+, show consistent performance improvement compared to the original settings.

Comparisons of different MSST variants are also illustrated in Table I. The weakly related task prompts of Variant-1 result in its relatively weak performance compared to other variants. The performance of Variant-2 depends on its teacher, and different LVLMS teachers perform differently on different datasets. On CHAMELEON, Variant-2-B excels Variant-2-I. Variant-2-L performs well on all datasets except on COD10K where it falls behind Variant-2-I. Combining multiple LVLMS with CLIP, Variant-3 achieves reasonably outstanding performance on all four datasets.

C. Ablation Study

We conduct ablation experiments on CHAMELEON, CAMO, and COD10K to analyze the effectiveness of each key component. For each image, we set the text input to Stable Diffusion to “a photo of a camouflaged animal” as the baseline model. If not specified, the following ablation experiments are conducted based on Variant-2-L.

1) *Effectiveness of MSST*: As illustrated in Table II, compared to the baseline, for CHAMELEON, CAMO, and COD10K, the introduction of MSST results in an average metric improvement of 4.2%, 0.7%, and 6.9%, respectively. Furthermore, attention maps with multi time steps and resolutions also contribute to UpGen. Iterative inference of SAM achieves limited results because initial prompts from attention maps capture object localization well enough.

TABLE II
ABLATION EXPERIMENTS OF VARIANTS WITH UPGEN ON THREE COD DATASETS

MSST	Generative Models		SAM	CHAMELEON				CAMO				COD10K			
	multi-time-step	multi-scale	iterative inference	$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^\omega \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^\omega \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^\omega \uparrow$	$M \downarrow$
✓				0.765	0.819	0.679	0.089	0.748	0.797	0.674	0.113	0.711	0.740	0.561	0.110
✓	✓			0.798	0.870	0.728	0.054	0.751	0.802	0.688	0.107	0.770	0.817	0.654	0.065
✓	✓	✓		0.797	0.861	0.722	0.058	0.764	0.817	0.703	0.101	0.774	0.822	0.660	0.065
✓	✓			0.825	0.883	0.763	0.047	0.785	0.833	0.729	0.093	0.801	0.850	0.703	0.054
✓	✓	✓	✓	0.820	0.874	0.757	0.048	0.793	0.842	0.737	0.089	0.802	0.852	0.703	0.053

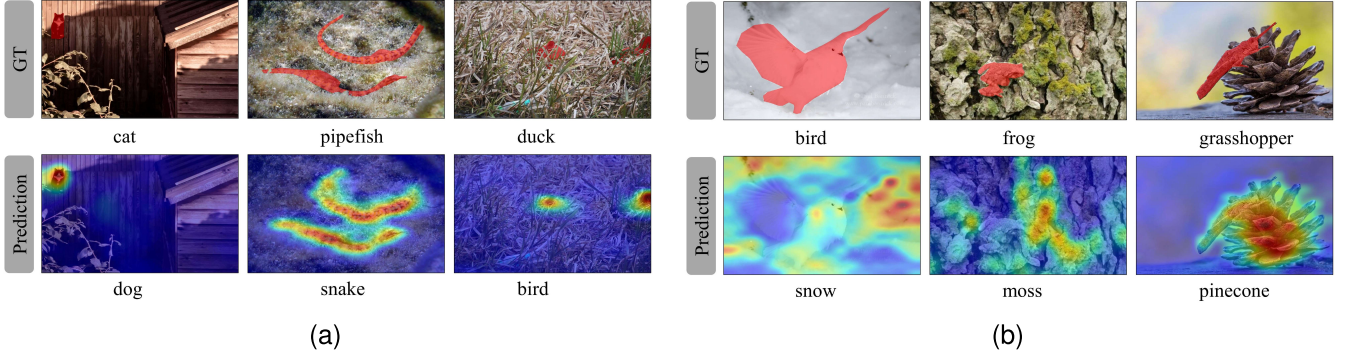


Fig. 7. (a) The diffusion model exhibits a relatively good tolerance to the wrong categories from MSST; (b) When MSST generates irrelevant environment categories, the attention maps fail to locate targets.

TABLE III
COMPARISON OF CATEGORY PREDICTION ACCURACY OF MSST VARIANT-3 WITH STUDENTS ON COD10K

	STU.1	STU.2	STU.3	Variant-1
ACC	60.82	57.88	52.96	61.16

TABLE IV
COMPARISON OF CATEGORY PREDICTION ACCURACY OF MSST VARIANT-3 WITH STUDENTS ON COD10K

	STU.1	STU.2	STU.3	Variant-3
ACC	58.93	64.10	65.39	70.91

TABLE V
COMPARISON OF CATEGORY PREDICTION ACCURACY OF MSST VARIANT-2 WITH STUDENTS ON COD10K

		STU.1	STU.2	STU.3	STU.4	STU.5	STU.6	Variant-2
ACC	Variant-2-B	58.43	56.39	57.48	55.00	57.63	57.68	58.93
	Variant-2-I	59.82	66.38	63.45	62.31	64.89	63.85	64.10
	Variant-2-L	61.91	65.59	63.10	64.74	65.24	62.71	65.39

TABLE VI
EXPERIMENTS ON EFFECTIVENESS OF GENERATIVE PROMPTS ON OPEN-VOCABULARY (OVCAMO) AND REFERRING (R2C7K) CAMOUFLAGED OBJECT DETECTION. THE BEST RESULTS ARE BOLD TO HIGHLIGHT

Method	Venue	Sup.	OVCamo (3,770 images)				R2C7K (1,987 images)			
			$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^\omega \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^\omega \uparrow$	$M \downarrow$
GenSAM	AAAI24	TF	0.591	0.652	0.360	0.137	0.602	0.681	0.398	0.149
Grounded-SAM	arXiv24	TF	0.779	0.816	0.655	0.078	0.804	0.850	0.712	0.066
UpGen	-	TF	0.785	0.829	0.666	0.069	0.807	0.858	0.712	0.050

It is notable that the diffusion model may exhibit a relatively good tolerance to hallucinations. As depicted in Fig. 7(a), the

diffusion model can accurately localize camouflaged objects even when MSST outputs incorrect categories. We attribute this capability to the similarity shared between the categories in both textual and visual embedding spaces. To further illustrate the effectiveness of MSST in mitigating hallucinations, we compared MSST-generated categories with ground truth (GT) categories directly. Our evaluations were conducted on COD10K-TEST (2,026 images), where each image is associated with a category label. An output is deemed correct if it matches the GT category or if they are synonymous (e.g., “cheetah” and “leopard”). Accuracy was employed as the evaluation metric. We compared the three MSST variants with their respective students, as presented in Tables III, IV, and V. The results indicate that MSST variants either outperform or perform comparably to the best-performing students, demonstrating the effectiveness of MSST in alleviating hallucinations. It also shows that when student-level prior is not available, MSST ensures that the lower bound of the whole framework is at least the level of the best students and avoids the influence of badly performing students.

2) *Effectiveness of Generative Prompts*: To evaluate the effectiveness of generative prompts based on Stable Diffusion, we further utilize Grounded-SAM [96] for segmentation, conditioned on text prompts derived from either ground-truth categories or our MSST. Grounded-SAM is designed for open-vocabulary segmentation, where text prompts are first processed by GroundingDINO [97] to generate bounding boxes, which are subsequently used to guide SAM for segmentation. Given the same category inputs, the primary distinction between Grounded-SAM and UpGen lies in their respective prompt generation mechanisms: GroundingDINO-based versus diffusion-based. For text prompts based on ground-truth

TABLE VII
ABLATION EXPERIMENTS ON EFFECTIVENESS OF GENERATIVE PROMPTS BASED ON DIFFUSION MODELS. THE TOP THREE
RESULTS ARE HIGHLIGHTED IN RED, GREEN AND BLUE

Method	Venue	Sup.	CHAMELEON				CAMO				COD10K				NC4K			
			$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^\omega \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^\omega \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^\omega \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^\omega \uparrow$	$M \downarrow$
Grounded-SAM	arXiv24	TF	0.781	0.821	0.707	0.098	0.712	0.734	0.645	0.152	0.799	0.844	0.706	0.068	0.808	0.845	0.747	0.084
MSST Variant-1 (Ours)	-	TF	0.805	0.851	0.731	0.061	0.774	0.832	0.717	0.095	0.794	0.841	0.693	0.062	0.827	0.873	0.772	0.063
MSST Variant-2-B (Ours)	-	TF	0.814	0.870	0.750	0.057	0.760	0.818	0.690	0.105	0.782	0.827	0.674	0.070	0.818	0.862	0.757	0.070
MSST Variant-2-I (Ours)	-	TF	0.806	0.865	0.738	0.059	0.768	0.816	0.701	0.101	0.804	0.855	0.707	0.052	0.836	0.883	0.785	0.057
MSST Variant-2-L (Ours)	-	TF	0.820	0.874	0.757	0.048	0.793	0.842	0.737	0.089	0.802	0.852	0.703	0.053	0.836	0.883	0.785	0.056
MSST Variant-3 (Ours)	-	TF	0.817	0.875	0.747	0.047	0.779	0.833	0.718	0.091	0.806	0.856	0.708	0.051	0.838	0.887	0.788	0.054

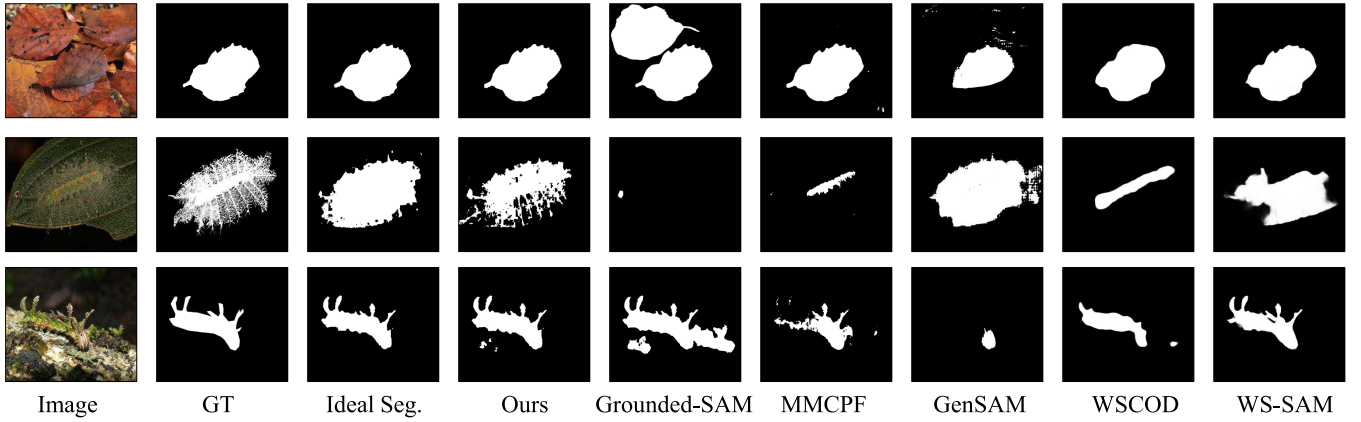


Fig. 8. Visual comparison of UpGen, IS, Grounded-SAM, MMC PF, GenSAM, WSCOD and WS-SAM.

categories, we conduct experiments on two tasks involving known object classes—open-vocabulary [98] and referring [99] camouflaged object detection. As shown in Table VI, our method demonstrates consistent performance gains. When using prompts generated by our MSST, as illustrated in Table VII, all variants of UpGen surpass Grounded-SAM across most evaluation metrics.

To further evaluate the validation of Generative Prompts, we adopt Ideal Segmentation (IS) for visual comparison with our method. IS obtains bounding boxes and points directly from the ground truth for segmentation, and the results could be considered as the upper bound of our method [25]. Fig. 8 shows the visual comparison of UpGen, IS, Grounded-SAM, training-free methods and weakly-supervised methods. Compared to these methods, our method is more capable of localizing objects for segmentation, thus avoiding misidentifying irrelevant surrounding objects as targets.

D. Further Analysis

1) *Sensitivity of Hyper-Parameter*: We analyze the sensitivity of four hyper-parameters, including image-level threshold (ILT), bbx-level threshold (BLT), number of sampling points (NSP), and number of SAM inference iterations (NIT) on CHAMELEON, CAMO, COD10K and NC4K. Following [51], we adopt score = $S_\alpha + E_\phi + F_\beta^\omega + (1 - M)$ to indicate the overall performance. The initial settings for ILT, BLT, NSP, and NIT are 0.3, 0.9, 1, and 2, respectively.

As shown in Fig. 9, UpGen is generally robust to BLT, NSP, and NIT. A higher BLT means that the prompts input into SAM have a higher confidence of containing camouflaged

objects, which can improve precision. However, increasing BLT may cause some less salient objects to be overlooked, thus reducing recall. For NSP, our method shows minimal fluctuation when the number of sampling points is small. However, as NSP increases, our method is more likely to sample from uncertain regions where objects are often seamlessly blended into their surroundings, which can confuse SAM and lead to incorrect segmentation results. Since the initial prompts provided to SAM already demonstrate strong localization of camouflaged objects, our method typically requires only one inference for satisfactory results. Increasing NIT does not significantly affect the performance of the model. ILT demonstrates a substantial influence on UpGen: minimal ILT may result in background noise, whereas excessive ILT could contribute to incomplete localization.

2) *Different Types of Prompts*: In this paper, for each instance, we select a bounding box along with a pair of positive and negative points from the attention map to serve as prompts for SAM. We investigate various types of prompts: points alone, and the combination of points and bounding boxes.

We first examine the scenario in which only point prompts are used. Two strategies are compared: using solely positive points versus using an equal number of positive and negative points. We do not consider the negative-point-prompt-only strategy, as it offers limited insights into camouflaged objects. As shown in Fig. 10(a), across all settings of sampling points, the combination of positive and negative points consistently outperforms the use of positive points alone. This is likely due to the inherent similarity between camouflaged objects and their surrounding environments. When only positive points

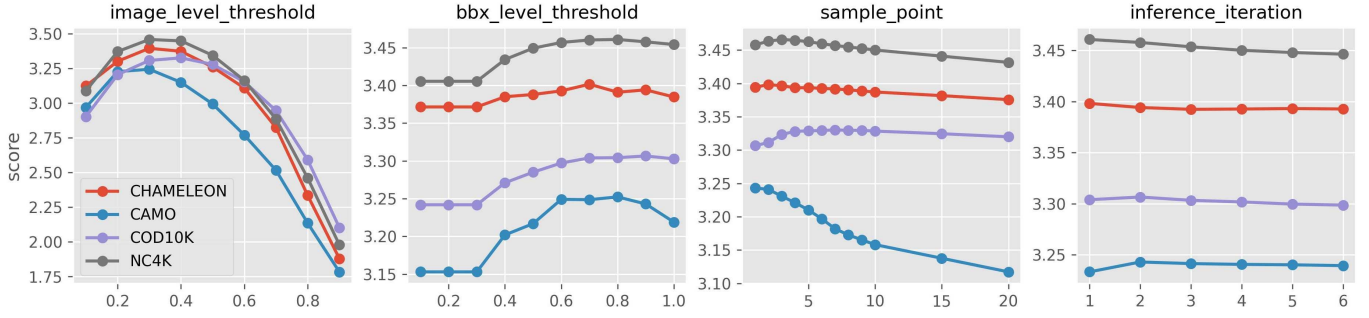


Fig. 9. Scores on four datasets with different hyper-parameter settings.

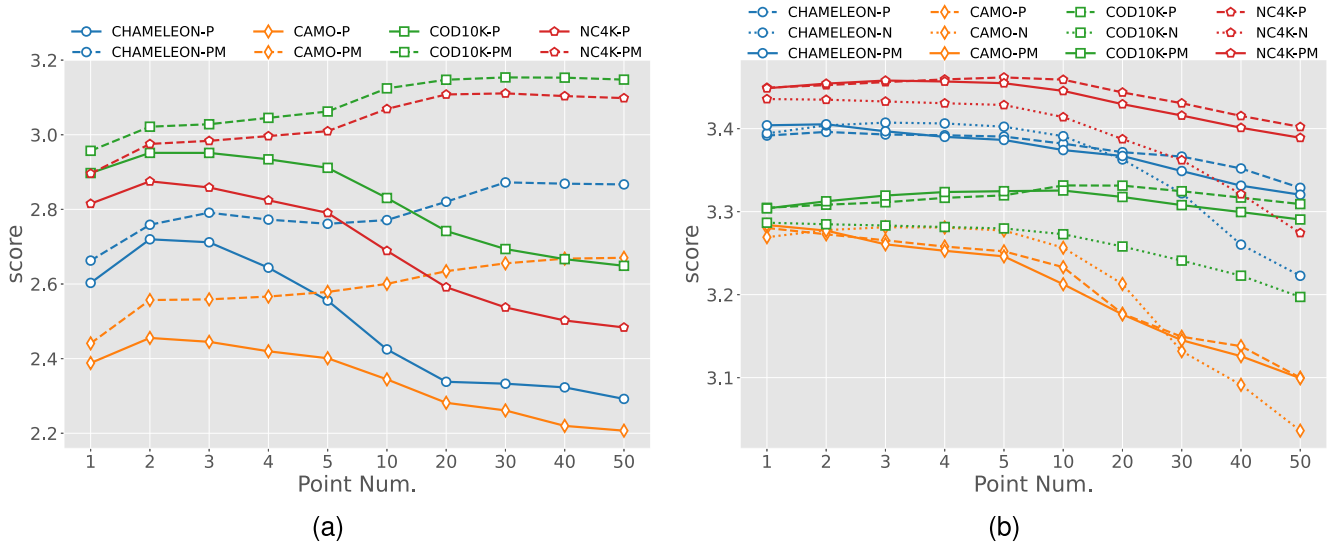


Fig. 10. (a) Ablation experiments on numbers and types of point prompts. “P” denotes positive point prompts while “PM” denotes both positive and negative point prompts; (b) Ablation experiments on mixed prompts. “P”: positive point and bounding-box prompts; “N”: negative point and bounding-box prompts; “PM”: both positive and negative point and bounding-box prompts.

are used, SAM may mistakenly segment background regions as part of the object, particularly in areas where object and background boundaries are ambiguous. Incorporating negative points helps to better distinguish the target from the background, guiding SAM toward more accurate segmentation, especially in visually confusing regions.

Moreover, under the positive-point-only strategy, we observe that increasing the number of sampled points actually degrades performance across all datasets. This can be attributed to the increased likelihood of selecting positive points in regions where the visual characteristics of the object and background are nearly indistinguishable, thus elevating the risk of mis-segmentation. In contrast, the inclusion of negative points mitigates this effect and establishes a positive correlation between model performance and the number of sampled points, within a reasonable range.

We then explore the scenario in which point prompts are combined with bounding boxes. This includes three configurations: boxes with positive points, boxes with negative points, and boxes with both positive and negative points. As shown in Fig. 10, all box-involving strategies significantly outperform the best-performing point-only

configuration across all datasets. While high-confidence point prompts provide localized and salient cues about camouflaged objects, bounding boxes contribute valuable global spatial context. A bounding box not only indicates the approximate location of the target but also implicitly highlights regions where the target is absent—an essential factor when segmenting camouflaged objects that closely resemble their surroundings.

In Fig. 10(b), building on the use of box prompts, we investigate the impact of varying the types and quantities of point prompts on model performance. When only a few points are used, incorporating positive, negative, or both types yields comparable results, with slight performance gains. However, as the number of sampled points increases, performance begins to deteriorate, which we attribute to the introduction of noisy or ambiguous points. Nevertheless, due to the accurate object localization provided by bounding boxes, the model consistently outperforms the best point-only configuration—even under suboptimal conditions. Based on these findings, we adopt a prompt strategy that combines a bounding box with a pair of positive and negative points for SAM throughout this paper.

TABLE VIII
ABLATION EXPERIMENTS OF DIFFERENT SAM VARIANTS

Method	SAM	Backbone	CHAMELEON				CAMO				COD10K				NC4K			
			$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^\omega \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^\omega \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^\omega \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^\omega \uparrow$	$M \downarrow$
MMCPF	HQ-SAM	VIT-H	-	-	-	-	0.749	0.820	0.680	0.101	0.733	0.803	0.592	0.066	0.767	0.826	0.681	0.083
GenSAM	SAM	VIT-H	0.767	0.827	0.673	0.075	0.738	0.803	0.674	0.106	0.773	0.832	0.667	0.065	0.810	0.866	0.751	0.065
Ours	SAM	VIT-B	0.784	0.851	0.709	0.067	0.728	0.784	0.665	0.122	0.759	0.815	0.652	0.072	0.795	0.852	0.738	0.074
		VIT-L	0.812	0.877	0.754	0.062	0.760	0.806	0.704	0.112	0.785	0.835	0.691	0.068	0.825	0.874	0.778	0.065
		VIT-H	0.814	0.876	0.758	0.057	0.763	0.818	0.705	0.103	0.794	0.847	0.700	0.058	0.826	0.876	0.781	0.061
	HQ-SAM	VIT-B	0.771	0.831	0.689	0.063	0.738	0.790	0.668	0.109	0.776	0.834	0.669	0.058	0.805	0.858	0.745	0.066
		VIT-L	0.813	0.870	0.751	0.051	0.771	0.822	0.709	0.097	0.800	0.850	0.702	0.054	0.832	0.881	0.781	0.057
		VIT-H	0.820	0.874	0.757	0.048	0.793	0.842	0.737	0.089	0.802	0.852	0.703	0.053	0.836	0.883	0.785	0.056

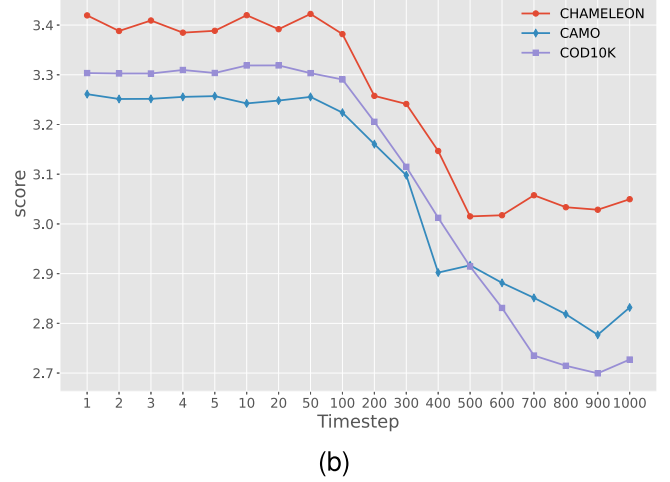
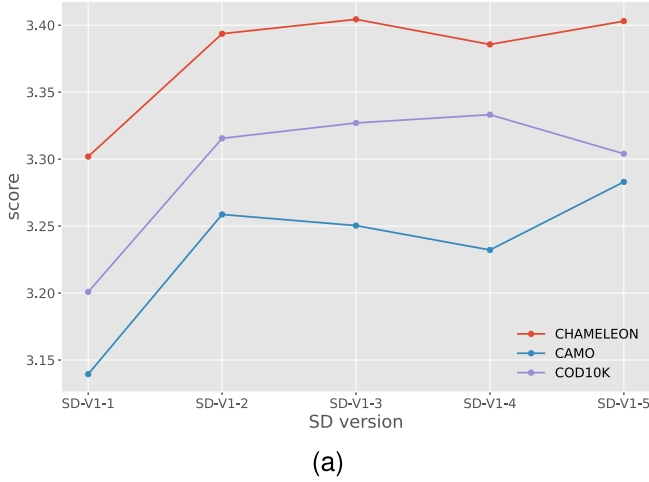


Fig. 11. (a) Ablation experiments of different versions of Stable Diffusion; (b) Ablation experiments on time steps in Stable Diffusion.

3) *Different Pre-Trained Models*: We adopt Stable Diffusion V1-5 [38] and HQ-SAM VIT-H [17] as the generative model and SAM, respectively. We compare two SAM variants: vanilla SAM [15] and HQ-SAM [17]. HQ-SAM could generate finer segmentation masks compared to SAM. In addition, we explore three different visual backbones for each SAM variant. It could be seen from Table VIII that 1) HQ-SAM outperforms SAM on the whole, 2) larger vision backbones perform better, and 3) our method with any SAM variant and vision backbones outperforms or equals other methods.

In addition, We conduct experiments on five versions of Stable Diffusion, including v1-1, v1-2, v1-3, v1-4, v1-5. Higher versions correspond to longer training time and therefore better quality of generation. As indicated in Fig. 11(a), in general, higher versions of Stable Diffusion are more capable of enhancing interactions between images and categories, thus improving segmentation performance. As its name suggests, UpGen embodies a continually evolving architecture. From above observations, it could be seen that our framework evolves alongside advancements in member models without reaching performance saturation.

4) *Different Time Steps*: We conducted ablation experiments to investigate the effect of different time steps on model performance. As shown in Fig. 11(b), the segmentation performance remains relatively stable when the time step is within the range of 1 to 100. Within this interval, selecting any time step yields satisfactory results. However, when the

number of steps exceeds 100, the diffusion model—without fine-tuning—begins to produce distorted attention maps due to increased noise variance, resulting in a noticeable decline in performance. In this study, we select attention maps from time steps 1, 50, and 100 to aggregate information across a diverse range of temporal stages. This choice enriches the representation by capturing complementary features while maintaining wide intervals to minimize inference overhead. Although other combinations may offer marginal gains, our selected configuration effectively balances performance and efficiency, capturing both class-specific distinctions and the localization of camouflaged objects.

5) *Inference Time*: We compare the inference time of UpGen with the most related training-free method, GenSAM [22]. The inference time (second) is **1.66** for UpGen and **20.20** for GenSAM, respectively. Compared to our UpGen, it takes GenSAM six iterations of inference to obtain satisfactory segmentation.

6) *Generalization of UpGen*: With minor task-related modifications to the prompts of MSST, UpGen could generalize to other applications. To verify this, we integrate UpGen to Salient Object Detection (SOD), Transparent Object Segmentation (TOS) and Marine Animal Segmentation (MAS). We directly replace the task-related prompt words, i.e., “camouflaged” is replaced by “salient” for SOD, “transparent” for TOS and “marine” for MAS, respectively. We compare UpGen with Grounded-SAM on seven widely used SOD

TABLE IX
GENERALIZATION TO SOD, TOS AND MAS. “*”: TOS DATASET. “#”: MAS DATASETS

Method	Venue	Sup.	MSRA-B				ECSSD				PASCAL-S				DUTS-TE			
			$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^\omega \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^\omega \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^\omega \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^\omega \uparrow$	$M \downarrow$
Grounded-SAM	arXiv24	TF	0.663	0.670	0.618	0.252	0.677	0.685	0.632	0.237	0.714	0.751	0.697	0.189	0.577	0.576	0.470	0.309
Ours	-	TF	0.852	0.888	0.811	0.070	0.874	0.907	0.842	0.055	0.766	0.822	0.758	0.126	0.782	0.813	0.689	0.092

Method	Venue	Sup.	HKU-IS				DUT-OMRON				SOD				Trans10K*			
			$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^\omega \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^\omega \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^\omega \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^\omega \uparrow$	$M \downarrow$
Grounded-SAM	arXiv24	TF	0.696	0.707	0.645	0.218	0.571	0.570	0.442	0.298	0.645	0.673	0.622	0.248	0.725	0.768	0.674	0.149
Ours	-	TF	0.858	0.898	0.821	0.055	0.738	0.772	0.613	0.112	0.786	0.834	0.772	0.103	0.636	0.709	0.570	0.209

Method	Venue	Sup.	MAS3K#				USOD10K#				RUWI#				UFO120#			
			$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^\omega \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^\omega \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^\omega \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^\omega \uparrow$	$M \downarrow$
Grounded-SAM	arXiv24	TF	0.811	0.845	0.722	0.072	0.814	0.845	0.750	0.074	0.551	0.561	0.451	0.275	0.742	0.778	0.695	0.128
Ours	-	TF	0.850	0.885	0.789	0.047	0.855	0.893	0.818	0.053	0.645	0.661	0.603	0.215	0.756	0.789	0.733	0.136

TABLE X
COMPARISON OF UPGen WITH IDEAL SEGMENTATION

Method	Venue	Sup.	CHAMELEON				CAMO				COD10K				NC4K			
			$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^\omega \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^\omega \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^\omega \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^\omega \uparrow$	$M \downarrow$
Ideal Segmentation	-	-	0.883	0.932	0.858	0.032	0.880	0.941	0.874	0.040	0.902	0.962	0.874	0.018	0.901	0.954	0.885	0.027
Ours(Mean of All Variants)	-	TF	0.812	0.867	0.745	0.054	0.775	0.828	0.713	0.096	0.798	0.846	0.697	0.058	0.831	0.878	0.777	0.060
Room for Improvement	-	-	7.1%	6.5%	11.3%	2.2%	10.5%	11.3%	16.1%	5.6%	10.4%	11.6%	17.7%	4.0%	7.0%	7.6%	10.8%	3.3%

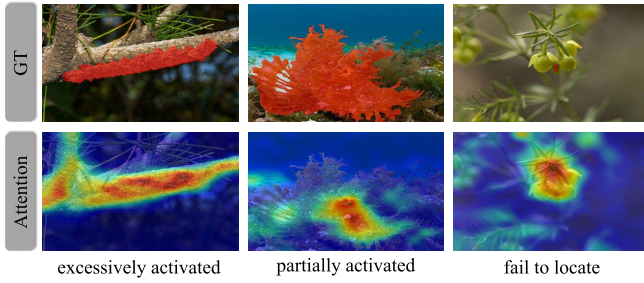


Fig. 12. Stable Diffusion may struggle to align images with text under some challenging cases.

datasets (i.e., MSRA-B [100], ECSSD [101], DUT-OMRON [102], PASCAL-S [103], HKU-IS [104], DUTS [105], and SOD [106]), a TOS dataset (Trans10K [42]), and four MAS datasets (i.e., MAS3K [107], USOD10K [108], RUWI [109], and UFO120 [110]). The same metrics are derived following COD. As shown in Table IX, UpGen substantially surpasses Grounded-SAM on all seven SOD datasets and four MAS datasets. However, it struggles with TOS. We’ll discuss this later.

7) *Room for Improvement*: We evaluate the Ideal Segmentation metrics across four COD datasets, as presented in Table X. Although our approach achieves state-of-the-art (SOTA) performance, there is still considerable room for improvement in UpGen. Several factors may contribute to performance degradation: (1) Rare words from MSST, such as *aardvark*, *kookaburra*, and *pronghorn*, may introduce ambiguity for Stable Diffusion; (2) In the absence of fine-tuning on COD datasets, Stable Diffusion may encounter difficulties in aligning image features with textual descriptions under challenging conditions. As shown in Fig. 12, even when MSST

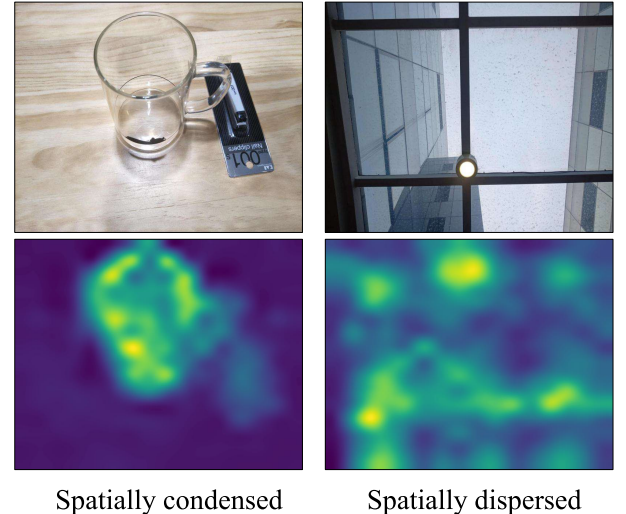


Fig. 13. Attention maps from transparent object segmentation.

correctly identifies the target categories, Stable Diffusion may either excessively activate or partially activate the attention maps, or fail to localize the targets entirely; (3) Due to the varying levels of camouflage across datasets, using uniform hyper-parameters can lead to inconsistent outcomes; (4) When MSST generates unrelated environmental categories, Stable Diffusion tends to focus on the background rather than the camouflaged subjects, as illustrated in Fig. 7(b).

Additionally, since UpGen is built upon Stable Diffusion, it naturally inherits the capability to segment spatially condensed objects. However, for spatially dispersed elements—such as windows, doors, or glass in transparent object segmentation—the attention maps produced by Stable

Diffusion become diffused throughout the image (see Fig. 13), resulting in noisy prompts. These limitations point to promising directions for future research.

V. CONCLUSION

In this paper, we propose to utilize generative models as “catalysts” to facilitate a promising “chemical reaction” between foundation models and COD in a training-free manner. We first propose the hallucination-mitigation framework, MSST, for automatically obtaining categories. Based on this, we leverage generative models to generate more fine-grained and semantic-rich prompts, which are used to prompt SAM for segmentation. Extensive experiments demonstrate that UpGen achieves better results compared to other SOTA methods with generative prompts, and could generalize to other tasks with minor modifications. In the future, we will explore more advanced generative models and large vision-language models for more complicated camouflaged object detection.

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